

Customer Churn Business Analysis

Executive Summary

Project Overview

This project analyzes customer churn data to identify important factors associated with customers leaving the company. The analysis was conducted using 500 customer records and included data cleaning, exploratory data analysis, statistical testing, and business recommendations.

Key Findings

- Overall customer churn rate was **10.6%**.
- Customers with **month-to-month contracts** had the highest churn rate at **20.59%**.
- Customers who churned had an average tenure of **6.00**, compared to **40.15** for non-churned customers.
- The difference in customer tenure was statistically significant.
- Churned customers had higher average monthly charges (**129.77**) than non-churned customers (**111.72**).
- Monthly charges showed a statistically significant association with churn.
- Total charges did not show a statistically significant difference between churned and non-churned customers.

Business Recommendations

1. Focus on retaining new customers during their early months.
2. Encourage month-to-month customers to choose longer-term contracts.
3. Review pricing and service value for customers with higher monthly charges.
4. Investigate customer experience across different payment methods.
5. Develop a proactive churn monitoring system to identify high-risk customers.

Conclusion

The analysis indicates that customer tenure, contract type, and monthly charges are important factors associated with customer churn in this dataset. The business can use these findings to develop targeted retention strategies and reduce customer churn.